

L3: Connected Components in $G(n,p)$

Clearly, there is no guarantee that the $G(n,p)$ model will give us a connected network. If p is close to zero, the network may consist of many small components. So an important question is: **how large is the Largest Connected Component (LCC) of the $G(n,p)$ model?**

Here is an animation that shows a network with $n=1000$ nodes, as we increase the average node degree $\bar{k}$ (shown at the upper-left of the animation). Recall that the connection probability is approximately $p \approx \frac{\bar{k}}{n}$.

As you see, initially we have only small groups of connected nodes -- typically just 2-3 nodes in every connected component.

After the first 5-10 seconds of the animation however, we start seeing larger and larger connected components. Most of them do not include any loops -- they form tree topologies.

Gradually, however, as the average degree approaches the critical value of one, we start seeing some connected components that include loops.

Something interesting happens when the average degree exceeds one (*about 40 seconds after the start of the animation*): the largest connected component (LCC), which is identified with a different color than the rest of the nodes, starts covering a significant fraction of all network nodes. It starts becoming a **“giant component”**.

If you continue watching this animation until the end (*it takes about five minutes*), you will see that this giant component gradually changes color from dark blue to light blue to yellow to red -- the color “*temperature*” represents the fraction of nodes in the LCC. Eventually, all the nodes join the LCC when the average node degree is about 6 in this example.